

ROS-Integrated Multi-Utility Autonomous Line Follower Bot for Factory Floor Operations

A Course Project Report

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CERTIFICATE

This is to certify that the report entitled ROS-Integrated Multi-Utility Autonomous Line Follower Bot for Factory Floor Operations submitted by **ASWATHY S** (TVE24ECRA02), **NITHIK PRABHAKAR** (TVE24ECRA09), **SHARANSUNDAR S** (TVE24ECRA13) to the APJ Abdul Kalam Technological University in partial fulfillment of the M.Tech. degree in Robotics and Automation is a bonafide record of the course project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Abstract

This project focuses on the design and implementation of an autonomous line follower robotic system for industrial automation, addressing key challenges such as material transportation, operational efficiency, and safety in factory environments. The system integrates a line-following robot equipped with ROS (Robot Operating System) to enable seamless navigation, real-time data processing, and adaptive task management. IR sensors are utilized for precise line tracking, while ultrasonic sensors ensure dynamic obstacle detection and smooth operation within industrial settings.

The robot's architecture employs Arduino for sensor and motor control, with ROSserial facilitating efficient communication between hardware and the ROS framework. This integration provides the system with intelligent decision-making capabilities, scalability, and optimized workflow execution. Key features include reliable path-following, obstacle avoidance, and automated package delivery, reducing reliance on manual labor and enhancing productivity.

Extensive testing in simulated and real-world environments demonstrated the system's effectiveness, reliability, and adaptability to complex industrial operations. By automating routine tasks and minimizing human intervention, this project underscores the significant role of robotics in industrial automation. The solution also lays a strong foundation for future advancements, including IoT integration, enhanced sensor technologies, and multi-robot collaboration, contributing to smarter and more efficient industrial processes.

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Chapter 1

Introduction

The continuous advancement of industrial automation technologies has redefined modern manufacturing and operational processes, enabling businesses to achieve higher levels of efficiency, precision, and productivity. Automation has become the cornerstone of industries striving to enhance throughput, reduce operational costs, and maintain consistent quality. This project centers on the development of an autonomous robotic system designed specifically for industrial applications. By integrating advanced capabilities such as line-following, obstacle avoidance, and automated task execution with the Robot Operating System (ROS), this multi-functional robot addresses the growing demands of automated operations in industrial settings, particularly in manufacturing, logistics, and material handling.

The backbone of this robotic system lies in its ability to automate the movement of materials and execute predefined tasks with minimal human intervention. One of the primary functionalities is its line-following capability, a fundamental requirement in many industrial environments where pathways and workflows are predefined for operational efficiency. This robot employs infrared (IR) sensors to detect and follow contrasting lines (typically black on a white surface), ensuring precise navigation along designated routes. Such precision is crucial in factory settings for applications such as conveyor belt synchronization, raw material transport, and finished goods delivery. The IR sensors provide continuous real-time feedback to the robot's control system, allowing it to make immediate adjustments to maintain alignment with the designated path.

In addition to line-following, the robot is equipped with ultrasonic sensors to detect and avoid obstacles dynamically. Industrial environments are often characterized by high activity, with workers, machines, and other equipment constantly moving. The ultrasonic sensors ensure the robot's safe navigation by measuring distances to objects in its surroundings and taking necessary corrective actions to avoid collisions. This dual capability of line-following and obstacle avoidance ensures uninterrupted operation, making the robot suitable for complex and dynamic factory environments where efficiency and safety are paramount.

The integration of the Robot Operating System (ROS) significantly enhances the capabilities and versatility of the robot. ROS provides a flexible and modular framework that allows seamless communication between the robot's hardware components, such as sensors, actuators, and controllers, and its software modules. This integration enables real-time data processing, task scheduling, and decision-making, which are essential for industrial automation. ROS simplifies the development and deployment of the system by offering prebuilt libraries and tools for critical tasks such as motion planning, sensor data fusion, and obstacle avoidance. Moreover, its modularity ensures that the robot remains adaptable to evolving industrial needs, allowing for future upgrades and scalability.

The robot's multi-tasking capability is another defining feature, enabling it to execute a variety of tasks such as material transportation, package delivery, and inventory management. The ROS-based task management system allows the robot to prioritize tasks based on operational requirements, ensuring efficient utilization of resources. For instance, the robot can autonomously transport items between workstations, avoiding bottlenecks and minimizing downtime. This reduces the need for manual intervention, allowing human workers to focus on higher-value tasks and improving overall productivity on the factory floor.

The project also emphasizes the importance of system reliability and adaptability. To ensure its robustness, the robot underwent extensive testing in both simulated and real-world environments. The simulated environment allowed for controlled evaluations of its capabilities, including handling path deviations, responding to obstacles, and switching tasks seamlessly. These tests provided valuable insights into the robot's performance under various scenarios, enabling iterative improvements.

Once validated in simulations, the robot was deployed in a real-world industrial setting, where it demonstrated its ability to navigate factory floors, adhere to predefined routes, and execute package deliveries efficiently. The successful deployment highlighted the system's reliability, adaptability to changing conditions, and effectiveness in performing multiple tasks autonomously.

The integration of automation through this autonomous robot showcases the transformative potential of robotics in industrial settings. By addressing critical challenges such as material handling, workflow optimization, and operational safety, the robot serves as a versatile solution for modern factories. The use of ROS ensures that the system is not only effective in its current configuration but also capable of adapting to future advancements in industrial automation technologies. As industries continue to adopt automation to meet the demands of Industry 4.0, autonomous robots like this will play a crucial role in driving innovation, enhancing productivity, and enabling the evolution of smart factories.

This project serves as a testament to the immense potential of integrating autonomous robotics with industrial automation. It highlights the importance of leveraging advanced frameworks like ROS and modern sensor technologies to create systems that are efficient, scalable, and capable of revolutionizing traditional industrial processes.

Chapter 2

Design

The design of a multi-functional autonomous robot for industrial applications requires a thoughtful integration of hardware components and software tools to ensure reliable operation in dynamic environments. This project leverages ROS (Robot Operating System) alongside hardware platforms such as Arduino and Raspberry Pi to create a versatile and efficient robot capable of performing tasks such as line-following, obstacle avoidance, package deliveries. The use of simulation tools like Gazebo and RViz is crucial for testing, visualization, and real-time monitoring during development..

2.1 Hardware design and integration

The hardware design of the robot involves various critical components, including sensors, actuators, and the hardware platform. The main goal is to ensure that all these components work seamlessly with the ROS-running software to allow autonomous operation in industrial environments.

The robot hardware is carefully designed to ensure reliable performance in a variety of industrial tasks. It integrates several key components that work together to facilitate autonomous operation, including sensors, motors, actuators, and the underlying hardware platforms Arduino and Raspberry Pi.

Sensors

The primary function of the robot is to follow a line, a task accomplished using infrared (IR) sensors. The IR sensors detect the contrast between a black line and a light-colored surface, allowing the robot to track and follow the path. Sensors continuously provide real-time feedback to the robot control system, enabling it to adjust its position and ensure accurate navigation.

In addition, the robot is equipped with ultrasonic sensors to avoid obstacles. These sensors emit sound waves and measure the time it takes for the waves to reflect back after encountering an object. The ultrasonic sensors enable the robot to detect and avoid obstacles in its environment, ensuring smooth navigation without collisions. The integration of both IR and ultrasonic sensors allows the robot to perform line-following and obstacle avoidance tasks simultaneously, providing a highly effective navigation system for dynamic environments.

Motors and Actuators

The movement of the robot is powered by DC gear motors, which drive the wheels. The motors are controlled by Arduino Uno, which interfaces with the sensors and actuators to control the robot's speed, direction, and responsiveness. The Arduino processes sensor inputs and outputs control signals to adjust motor speed and direction, ensuring that the robot follows the line and avoids obstacles in real time.

Microcontroller Integration – Arduino Uno

The Arduino Uno acts as the microcontroller for low-level control, responsible for motor control and sensor integration. It reads data from the IR and ultrasonic sensors, processes the information using simple control algorithms, and sends commands to adjust the motors' speed and direction. The Arduino ensures precise control over the robot's movement, allowing it to stay on course while reacting to obstacles in its path. This makes Arduino a suitable choice for handling low-latency, time-critical operations such as motor control and sensor data reading.

Main Processing Unit – Raspberry Pi

The Raspberry Pi serves as the main processing unit, running the Robot Operating System (ROS) and handling high-level decision-making tasks. The Raspberry Pi communicates with the Arduino through serial communication, receiving sensor data and sending back control commands. It is responsible for managing complex algorithms for path planning, task management, and dynamic decision-making, such as prioritizing package deliveries.

The Raspberry Pi's role is critical in processing the sensor data, executing the task management system, and integrating all the functionalities required for the robot's autonomous operations. By running ROS, the Raspberry Pi enables the robot to coordinate multiple processes concurrently, such as line-following, obstacle avoidance, and task execution, while also supporting future expansion of the system.

2.2 Software framework and Task Management

ROS serves as the central framework for integrating the hardware components, processing sensor data, and managing tasks. ROS allows for modular development, enabling the robot to be programmed in a flexible and scalable way. The ROS framework divides the system into nodes, each responsible for specific tasks such as motion control, sensor data processing, and task management.

Line-Following Algorithm

The line-following algorithm is a core component of the robot's functionality. It utilizes the input from the IR sensors to detect the robot's position relative to the black line. When the robot drifts off the line, the algorithm calculates the necessary adjustments in motor speeds to steer it back onto the path. The communication between the Raspberry Pi and Arduino ensures that the line-following behavior is executed smoothly, with the Raspberry Pi processing the overall motion plan and the Arduino handling real-time motor control.

Obstacle Avoidance Algorithm

Obstacle avoidance is implemented through the data provided by the ultrasonic sensors. The Raspberry Pi uses the distance readings from these sensors to detect potential obstacles in the robot's path. When an obstacle is detected within a defined threshold, the robot adjusts its direction, either by stopping or rerouting to avoid the obstacle while maintaining its course on the line. The decision-making process is handled by the ROS-based task scheduler, ensuring real-time adjustments to prevent collisions without deviating from the mission at hand.

Task Management System

In addition to line-following and obstacle avoidance, the robot is designed to execute specific tasks such as package deliveries . The task management system prioritizes these tasks dynamically based on real-time data. The ROS framework facilitates the coordination of these tasks by dynamically adjusting the robot's behavior based on the system's inputs and priorities.

2.3 Simulation and Testing with Gazebo and RViz

Before deploying the robot in a real-world factory environment, extensive testing is performed using Gazebo and RViz, two powerful simulation and visualization tools provided by ROS.

Gazebo for Simulation

Gazebo provides a physics-based simulation environment where the robot's movement can be tested under controlled conditions. In this virtual environment, the robot's sensors, actuators, and algorithms are modeled to mimic real-world behavior. Gazebo allows for the simulation of various operational scenarios, such as detecting obstacles, following lines, and performing multi-tasking operations. This is particularly useful in identifying potential issues with the robot's performance before actual deployment. It also supports testing in various environments, including factory floors with dynamic obstacles, ensuring the robot is prepared for real-world challenges.

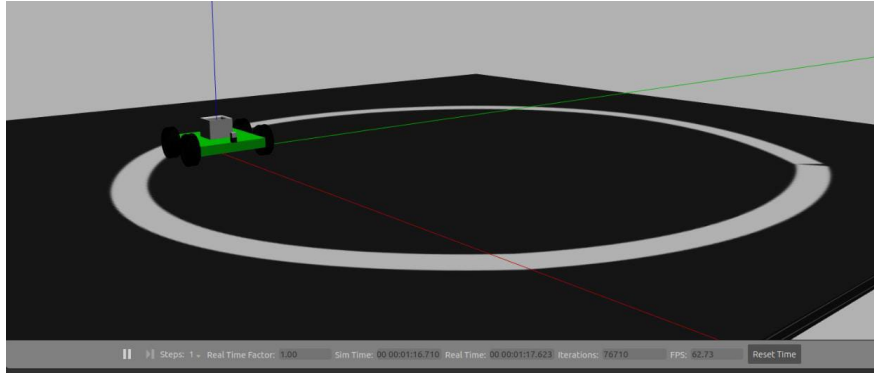


Figure 2.1: Gazebo Simulation of line follower model

RViz for Visualization

RViz is used to visualize the robot's state, the environment, and the sensor data in real-time. By displaying the robot's trajectory, sensor readings, and obstacles, RViz helps developers monitor the robot's performance during simulation and real-world testing. RViz also provides a platform for debugging, allowing developers to view how the robot perceives its environment and make adjustments to improve its behavior.

2.4 Working principle

The working principle of the line follower robot with obstacle avoidance and package delivery in a factory setting is based on the integration of IR sensors for navigation and ultrasonic sensors for obstacle collision avoidance. The robot operates without using a PID controller or a finite state machine (FSM), relying instead on straightforward sensor-based logic and a motor drive powered by 12v Lithium-ion battery pack for mobility.

The IR sensors, mounted on the underside of the robot, detect the contrast between the guiding line (e.g., a black line on a white floor or vice versa) and the surrounding floor. When the robot deviates from the line, the IR sensors send signals to the control system, which adjusts the motor speeds to realign the robot with the line. The control system uses a simple logic-based approach where motor speeds are adjusted based on whether the left or right sensors detect the line, ensuring the robot follows the path effectively.

For obstacle avoidance, the ultrasonic sensors measure the distance to objects in

the robot's path by emitting sound waves and calculating the time taken for the echo to return. If an obstacle is detected within a predefined threshold distance, the robot halts and executes a collision avoidance maneuver. The maneuver typically involves predefined steps, such as turning left or right, moving forward to bypass the obstacle, and realigning with the guiding line once the path is clear.

The lead-acid battery provides the power required for the motors, motor drivers, and sensors, ensuring consistent operation throughout the robot's tasks. The motor driver interfaces between the control system and the motors, translating control signals into the necessary power and direction for the motors to move the robot.

During operation, the robot follows the designated line on the factory floor to navigate between pickup and delivery points. At the pickup point, a simple mechanical or servo-actuated mechanism collects the package. The robot then continues along the line to the delivery point, avoiding obstacles along the way. At the delivery location, the mechanism releases the package before the robot proceeds to the next task or returns to its starting position.

This approach ensures reliable operation with minimal complexity, making the robot well-suited for factory floor applications where cost-effectiveness and straight-forward functionality are priorities

Chapter 3

Implementation in industrial automation

3.1 Introduction

Industrial automation has become a pivotal factor in transforming manufacturing and operational processes, enabling businesses to achieve unprecedented efficiency, accuracy, and scalability. This chapter outlines the implementation of a robotic system designed for industrial automation, focusing on integrating line-following capabilities, obstacle avoidance, and task execution using cutting-edge technologies such as the Robot Operating System (ROS), infrared (IR) sensors, ultrasonic sensors, and Arduino microcontrollers. By addressing the challenges faced in modern industrial settings, this implementation highlights the potential of robotics in streamlining operations and reducing human intervention.

3.2 System Design

The system design for the industrial automation project focuses on a robust, reliable robotic platform capable of automating material transportation and enhancing operational efficiency in dynamic industrial environments. The core functionality of the robot is divided into several components, each playing a critical role in ensuring the robot performs its tasks autonomously and effectively.

The line-following mechanism is one of the foundational elements of the system. IR sensors are placed underneath the robot to detect and follow a black line, typically found on factory floors or conveyor belts. This allows the robot to navigate predetermined paths without manual input. The Arduino microcontroller processes the sensor data, interpreting the robot's position relative to the line and controlling its movement accordingly. DC motors, connected to a motor driver module, are responsible for moving the robot forward, turning, or halting, depending on the commands received from the Arduino.

The obstacle avoidance system is integrated to enhance the robot's adaptability in an industrial setting. Ultrasonic sensors are strategically placed on the robot to continuously measure the distance to any obstacles in its path. If an obstacle is detected within a predefined distance (e.g., 20 cm), the robot will pause its forward motion, assess the situation, and adjust its route to avoid a collision. The integration of the line-following and obstacle avoidance algorithms ensures that the robot can seamlessly switch between navigating along a line and avoiding obstacles, depending on the situation.

A key part of the system is the carrier assembly designed for material handling and package transportation. This carrier is securely mounted on the robot's chassis, providing a stable platform for carrying items such as packages, materials, or tools within a factory environment. The carrier is designed to accommodate various types of materials, with adjustable slots or compartments that can hold objects of different sizes and weights. The integration of the carrier with the robot's movement ensures that materials are transported without risk of shifting or falling during transit. Additionally, the carrier can be customized to suit specific industrial requirements, such as incorporating sensors or actuators for load sensing or automatic unloading.

The integration framework relies heavily on the Robot Operating System (ROS), which serves as the communication backbone between the robot's hardware and software. ROS allows for modular programming, enabling the robot to handle complex tasks such as real-time sensor data processing, decision-making, and control over the various components, including sensors, motors, and the carrier assembly. The use of ROSserial ensures that the Arduino microcontroller can efficiently communicate with the ROS system, providing real-time feedback on the robot's status and task execution.

To power the entire system, a 12v Lithium ion battery pack is employed, ensuring that the robot operates autonomously for extended periods without requiring frequent recharging. The battery powers the motors, sensors, microcontroller, and other electronic components, providing the necessary energy to maintain continuous operation on the factory floor.

This system design creates a versatile robotic platform that can automate tasks such as package delivery, material transportation, and route navigation in an industrial setting. With the integration of advanced sensors, a custom carrier for handling packages, and a flexible software framework via ROS, the system promises to significantly reduce human intervention while enhancing the productivity and efficiency of industrial operations.

3.3 Implementation process

The implementation process of the industrial automation system begins with configuring the hardware components that form the backbone of the robotic platform. A stable chassis is assembled to support the sensors, motors, and the carrier assembly, which is a key component for material handling within the industrial environment. The carrier system is specifically designed to transport materials or packages securely across the factory floor. It is mounted on the robot's chassis and features compartments or adjustable platforms that can hold a variety of package sizes and weights, ensuring that materials are stable and safe during transport. This carrier system is essential for automating material handling, reducing the need for human intervention and increasing operational efficiency. The robot's IR sensors are placed beneath the chassis to detect and follow a black line, enabling the robot to navigate predefined paths. The ultrasonic sensor is positioned at the front to measure distances to obstacles, helping the robot avoid collisions. The DC motors, controlled via an H-bridge motor driver, provide precise movement for the robot, while the carrier ensures safe and stable transportation of materials between different points on the factory floor.

Next, the software development process begins, utilizing the Robot Operating System (ROS) as the central framework for communication between hardware and software components. ROS enables modular programming, allowing the robot to

perform tasks such as data processing, task management, and sensor control. The Arduino microcontroller, which interfaces with ROS through ROSserial, processes sensor data and controls the robot's movements. The line-following algorithm is designed to interpret data from the IR sensors, ensuring the robot remains on track along its path. The obstacle avoidance algorithm ensures the robot detects obstacles using the ultrasonic sensor, halting or rerouting when necessary.

Once the hardware is configured and the software is developed, the system undergoes calibration and testing. The IR and ultrasonic sensors are calibrated to function reliably under varying lighting and surface conditions. The motors are tested for precise movement control, and the system is tested in real-world environments, simulating the actual conditions of the factory. The robot, with its carrier system, is evaluated for its ability to transport packages securely while navigating factory routes and avoiding obstacles. Final integration includes ensuring the robot can efficiently handle material transportation tasks, with the carrier system designed to accommodate various package sizes and weights. Once fully tested, the system is deployed for autonomous operation, enhancing productivity, reducing human labor, and streamlining material handling processes in industrial environments.

Conclusion

The successful implementation of the industrial automation system showcases the significant potential of robotics in modernizing material handling and improving operational efficiency in industrial settings. Through the integration of technologies like the Robot Operating System (ROS), IR sensors, ultrasonic sensors, and Arduino microcontrollers, the system is capable of autonomous navigation, obstacle avoidance, and efficient task execution. A critical aspect of this system is the carrier assembly, which plays an essential role in automating material handling. The carrier system ensures that materials and packages are securely transported across the factory floor, providing stability and preventing loss or damage during movement. This advancement minimizes the need for human intervention and reduces manual labor in the handling of materials, leading to increased productivity.

Chapter 4

Results

4.1 Observation

The line follower robot demonstrated essential functionalities such as obstacle avoidance and package delivery, showcasing its potential for factory floor automation. The obstacle avoidance system, powered by ultrasonic sensors, performed reliably by accurately detecting obstacles within the predefined threshold distance and executing precise maneuvers to navigate around them. This ensured smooth and safe operation, even in dynamic environments with potential obstructions.

The package handling mechanism functioned as expected, enabling the robot to pick up and deliver packages at designated locations. This feature highlights the robot's capability for automating material handling tasks in industrial settings. While occasional deviations in navigation affected the accuracy of reaching the pickup and drop-off points, the integration of this mechanism demonstrated the robot's overall effectiveness in logistics management.

Initially, the robot was designed to use a Raspberry Pi as the primary controller for managing navigation and task execution via ROS Noetic. However, due to hardware compatibility and performance issues, the Raspberry Pi could not be integrated successfully. As an alternative, `rosserial` was implemented to establish communication between a microcontroller (e.g., Arduino) and the ROS framework. This alternative setup allowed the robot to process sensor data, control motor movements, and execute tasks effectively, demonstrating the system's adaptability and ensuring that core

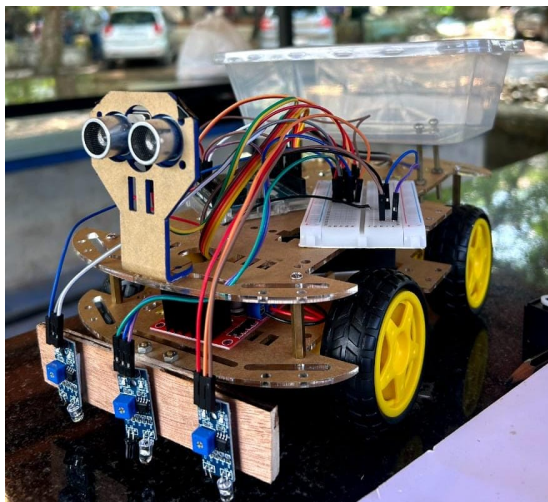


Figure 4.1: Final model of the line follower bot

functionalities were maintained despite the challenges.

In the line-following functionality, the robot encountered significant challenges, which impacted its navigation performance. Initially, a 5-channel IR array sensor was used to detect the guiding line. However, the robot failed to receive any readings from this sensor, likely due to calibration or hardware issues. As a result, the system was modified to use three individual IR sensors. While this solution enabled basic line detection, the reduced number of sensors limited the robot's ability to track the narrow line accurately. This, combined with the narrow width of the guiding line, led to frequent deviations from the path, affecting the robot's ability to navigate smoothly and complete tasks consistently.

In summary, the robot exhibited robust obstacle avoidance, functional package handling, and successful integration with ROS via roserial. While the Raspberry Pi and the 5-channel IR array sensor could not be utilized as planned, the alternative solutions allowed the robot to maintain its basic functionality. The challenges observed in line-following accuracy—due to the narrow guiding line and the use of fewer IR sensors—provide a valuable opportunity for improvement in future iterations. With enhancements such as a wider guiding line, optimized sensor placement, or advanced algorithms, the robot has the potential to become a highly reliable solution for automating operations in industrial environments.

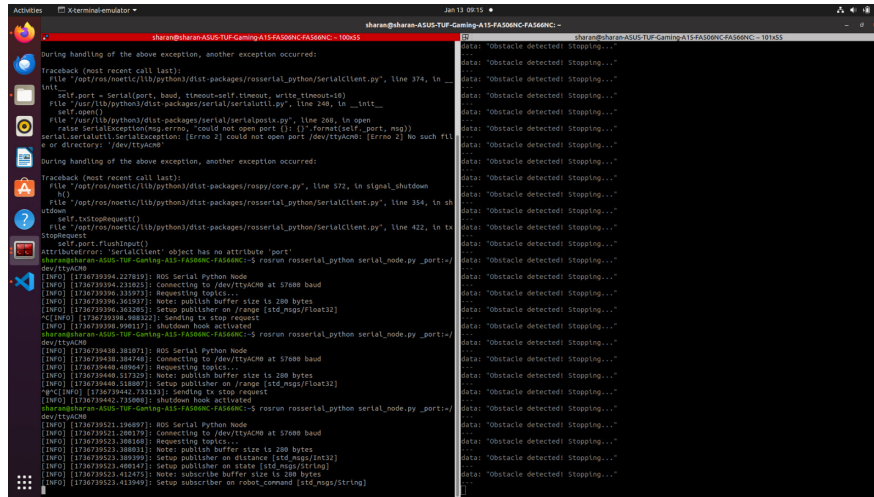
4.2 Result

The line follower robot demonstrated a strong foundation in key areas such as obstacle avoidance, package delivery, and system integration, though there were certain challenges that provided valuable insights for future improvements. The obstacle avoidance system, utilizing ultrasonic sensors, performed exceptionally well. The robot reliably detected obstacles and successfully executed maneuvers to avoid collisions, ensuring continuous, smooth operation even in dynamic environments. This aspect of the robot's performance met the design objectives for safety and path optimization, demonstrating robust functionality.

Similarly, the package handling mechanism worked as designed, enabling the robot to pick up and deliver packages to predefined locations. This capability showcased the robot's potential for automating material handling tasks on the factory floor. While the navigation issues did cause some inaccuracies in reaching specific pickup and drop-off points, the robot's overall ability to carry out the core logistics functions remained intact.

The line-following functionality did present some challenges, but these provided key learning opportunities for future optimization. Initially, the robot was equipped with a 5-channel IR array sensor, but it was unable to provide reliable readings, likely due to calibration or hardware issues. In response, the system was modified to use three individual IR sensors. While this adjustment enabled basic line detection, the narrow width of the guiding line and the limited sensor array led to occasional deviations from the intended path. This area can be further optimized in future versions, either through adjustments to sensor placement, expanding the number of sensors, or increasing the width of the guiding line, all of which would significantly improve the robot's line-following accuracy.

Regarding system integration, the original plan to utilize a Raspberry Pi was not feasible due to hardware compatibility issues. However, this challenge was effectively addressed by implementing `rosserial` for communication between the microcontroller and the ROS framework. This solution ensured that the robot could maintain communication between all components, process sensor data, and execute basic movement commands, allowing the robot to perform its tasks effectively despite the



The screenshot shows a terminal window with two panes. The left pane displays a Python traceback for a `SerialException` and an `AttributeError` related to the `SerialClient` object. The right pane shows a continuous stream of ROS log messages, including `INFO` messages about connecting to `/dev/ttyACM0` at 57600 baud, setting up publishers and subscribers, and sending stop requests. Interspersed among these are multiple `data: "Obstacle detected! Stopping..."` messages, indicating the robot's response to detected obstacles.

Figure 4.2: Simulation result

initial setback with the Raspberry Pi.

In conclusion, the robot demonstrated solid performance in several critical areas, including obstacle avoidance, package handling, and effective system integration through roserial. While the line-following accuracy and the Raspberry Pi integration presented challenges, these issues provided valuable insights for refining the design and enhancing the robot's capabilities. By addressing these areas of improvement, such as optimizing sensor configuration and expanding the robot's computational power, the system has significant potential to evolve into a highly effective solution for factory floor automation.

Chapter 5

Discussion

5.1 Future Additions

Line Following and Navigation Enhancements

One of the key areas of improvement in the line follower robot is its line following capability. Currently, the robot uses a narrow guiding line with a limited number of IR sensors, resulting in occasional deviations from the path. To enhance navigation, future versions of the robot could incorporate a wider sensor array, which would allow more accurate line detection and smoother path following. Furthermore, employing PID control algorithms would significantly improve the robot's ability to follow the line even with minor misalignments or irregularities in the path. Expanding the line width would also contribute to more stable and reliable line-following performance under varying conditions.

To further improve the robot's navigation system, the integration of additional sensors, such as LIDAR or enhanced infrared sensors, could be beneficial. These sensors would provide better obstacle detection and environmental awareness, allowing the robot to adapt more effectively to dynamic factory floor environments. Combining these with advanced path-planning algorithms would enable the robot to navigate more complex spaces with greater autonomy.

System Control and Computational Power

While the current control system, utilizing rosserial, has allowed basic functionality, the future of the robot would benefit from the integration of a more powerful computing unit. Incorporating a Raspberry Pi or similar single-board computer would provide the robot with increased processing power, allowing it to execute more sophisticated real-time decision-making algorithms. This would enable dynamic route planning, advanced obstacle avoidance, and more responsive system behavior, making the robot better suited for a wide range of unpredictable environments.

Further integration of wireless communication technologies, such as Wi-Fi or Bluetooth, would enable remote control, monitoring, and troubleshooting. This addition would allow operators to interact with the robot from a distance, making it easier to manage multiple robots or monitor their status in real time, improving overall workflow in factory operations.

Task Execution and Efficiency Enhancements

For better task execution, incorporating a vision system or camera-based navigation would significantly improve robot ability to interact with its environment. Using computer vision to detect objects, recognize package locations, and identify drop-off points would improve the robot's precision in handling packages. Such a system would be especially useful in environments where the layout frequently changes or where the task locations are dynamic.

Additionally, improving the robot's power system by replacing the current 12v Lithium-ion battery pack with a lithium-polymer battery pack would increase energy density and reduce weight. This upgrade would not only extend the robot's operational time, but also enhance its maneuverability. Further optimization of power consumption during idle states and low-power modes would contribute to better overall energy efficiency, extending battery life, and minimizing the need for frequent recharging.

5.2 Implementation Overview

During the course of the project, several challenges were encountered, particularly with integrating the Raspberry Pi as the primary control unit. Compatibility and performance issues with the Pi led to the decision to use rosserial for communication between the microcontroller and the ROS framework. This shift allowed the robot to continue functioning, although with more basic control capabilities compared to the initially intended setup.

In terms of the line-following mechanism, the initial use of a 5-channel IR array sensor failed to provide reliable readings, prompting the team to switch to a configuration using three individual IR sensors. While this modification allowed for basic line detection, the narrow guiding line and the reduced sensor array led to deviations from the path, limiting the robot's line-following accuracy.

Despite these challenges, the robot successfully demonstrated its obstacle avoidance functionality, where ultrasonic sensors were able to detect and avoid obstacles with high accuracy. The robot also performed well in its package handling tasks, picking up and delivering items to specified locations, though occasional navigation inaccuracies affected its precision in reaching target points.

5.3 Conclusion

The project demonstrated a functional prototype of a line-following robot with obstacle avoidance and package delivery capabilities, which has significant potential for industrial automation. While the robot showed solid performance in terms of obstacle avoidance and package handling, the line-following accuracy and system integration posed challenges that were addressed through adjustments in sensor configuration and control mechanisms. The use of `rosserial` proved to be a viable solution for communication between the microcontroller and ROS, though the integration of a more powerful computing system, such as a Raspberry Pi, remains a priority for future work.

Future developments in the project should focus on enhancing the line-following system, integrating more advanced sensors for navigation, and optimizing power efficiency through upgraded batteries. By addressing these key areas, the robot can evolve into a more robust solution capable of performing complex tasks reliably in dynamic environments. The combination of enhanced sensors, improved control systems, and longer operational times will allow the robot to better meet the demands of factory floor automation, paving the way for future applications in automated logistics and material handling.

Chapter 6

Code

ROS-Integrated Arduino Code for Line-Following Robot with Obstacle Detection and Multi-Wheel Motor Control

```
// Include necessary ROS libraries
#include <ros.h>
#include <std_msgs/Bool.h>
#include <std_msgs/String.h>
#include <std_msgs/Int32.h>

// Motor Driver Pins
const int in1 = 2; // Front-left motor IN1
const int in2 = 3; // Front-left motor IN2
const int in3 = 4; // Front-right motor IN1
const int in4 = 5; // Front-right motor IN2
const int in5 = 6; // Rear-left motor IN1
const int in6 = 7; // Rear-left motor IN2
const int in7 = 8; // Rear-right motor IN1
const int in8 = 11; // Rear-right motor IN2

// Ultrasonic Sensor Pins
const int trigPin = 12;
const int echoPin = 13;
```

```

// IR Sensor Pins
const int irLeft = A0; // Left IR sensor
const int irCenter = A1; // Center IR sensor
const int irRight = A2; // Right IR sensor

// Variables
long duration;
int distance;
bool obstacle_detected = false;
ros::NodeHandle nh;

// ROS Messages
std_msgs::Int32 distance_msg;
std_msgs::String state_msg;
std_msgs::String ir_values_msg;
std_msgs::String robot_command_msg;

// ROS Publishers
ros::Publisher distance_pub("distance", &distance_msg);
ros::Publisher state_pub("state", &state_msg);
ros::Publisher ir_values_pub("ir_values", &ir_values_msg);
ros::Publisher robot_command_pub("robot_command", &robot_command_msg);

// Commands received via ROS
String robot_state = "STOP";

// Function declarations
void commandCallback(const std_msgs::String &msg);
void followLine();
void moveForward();
void turnLeft();
void turnRight();
void stopMotors();
int measureDistance();

```

```

// ROS Subscriber callback
void commandCallback(const std_msgs::String &msg) {
    robot_state = String(msg.data);
    nh.loginfo(robot_state.c_str());

    // Publish the received command to the "robot_command" topic
    robot_command_msg.data = robot_state.c_str();
    robot_command_pub.publish(&robot_command_msg);
}

ros::Subscriber<std_msgs::String> command_sub("robot_command_input", &commandCallback);

void setup() {
    // Initialize ROS
    nh.initNode();
    nh.advertise(distance_pub);
    nh.advertise(state_pub);
    nh.advertise(ir_values_pub);
    nh.advertise(robot_command_pub);
    nh.subscribe(command_sub);

    // Motor control pins as output
    pinMode(in1, OUTPUT);
    pinMode(in2, OUTPUT);
    pinMode(in3, OUTPUT);
    pinMode(in4, OUTPUT);
    pinMode(in5, OUTPUT);
    pinMode(in6, OUTPUT);
    pinMode(in7, OUTPUT);
    pinMode(in8, OUTPUT);

    // Ultrasonic sensor pins
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
}

```

```

// IR sensor pins
pinMode(irLeft, INPUT);
pinMode(irCenter, INPUT);
pinMode(irRight, INPUT);
}

void loop() {
    // Measure distance
    distance = measureDistance();

    // Publish distance data
    distance_msg.data = distance;
    distance_pub.publish(&distance_msg);

    // Read and publish IR sensor values
    int left = digitalRead(irLeft);
    int center = digitalRead(irCenter);
    int right = digitalRead(irRight);

    char ir_values[50];
    sprintf(ir_values, "IR Left: %d, Center: %d, Right: %d", left, center, right);
    ir_values_msg.data = ir_values;
    ir_values_pub.publish(&ir_values_msg);

    // Obstacle avoidance
    if (distance > 0 && distance < 20) {
        stopMotors();
        state_msg.data = "Obstacle detected! Stopping...";
        state_pub.publish(&state_msg);
        obstacle_detected = true;
    } else {
        if (obstacle_detected) {
            state_msg.data = "Obstacle cleared: Resuming...";

```



```

    state_pub.publish(&state_msg);
    obstacle_detected = false;
}

if (robot_state == "FOLLOW_LINE") {
    followLine();
} else if (robot_state == "MOVE_FORWARD") {
    moveForward();
} else if (robot_state == "TURN_LEFT") {
    turnLeft();
} else if (robot_state == "TURN_RIGHT") {
    turnRight();
} else {
    stopMotors();
}
}

nh.spinOnce();
delay(50); // Small delay for stability
}

void followLine() {
    // Read IR sensor values
    int left = digitalRead(irLeft);
    int center = digitalRead(irCenter);
    int right = digitalRead(irRight);

    if (center == HIGH && left == LOW && right == LOW) {
        moveForward();
        state_msg.data = "Line detected: Moving forward";
    } else if (left == HIGH && center == LOW && right == LOW) {
        turnLeft();
        state_msg.data = "Line detected: Turning left";
    } else if (right == HIGH && center == LOW && left == LOW) {

```

```

        turnRight();
        state_msg.data = "Line detected: Turning right";
    } else {
        stopMotors();
        state_msg.data = "Line lost: Stopping";
    }
    state_pub.publish(&state_msg);
}

void moveForward() {
    // Front motors
    digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
    digitalWrite(in3, HIGH); digitalWrite(in4, LOW);
    digitalWrite(in5, HIGH); digitalWrite(in6, LOW);
    digitalWrite(in7, HIGH); digitalWrite(in8, LOW);
}

void turnLeft() {
    // Front motors
    digitalWrite(in1, LOW); digitalWrite(in2, HIGH);
    digitalWrite(in3, HIGH); digitalWrite(in4, LOW);
    digitalWrite(in5, LOW); digitalWrite(in6, HIGH);
    digitalWrite(in7, HIGH); digitalWrite(in8, LOW);
}

void turnRight() {
    // Front motors
    digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
    digitalWrite(in3, LOW); digitalWrite(in4, HIGH);
    digitalWrite(in5, HIGH); digitalWrite(in6, LOW);
    digitalWrite(in7, LOW); digitalWrite(in8, HIGH);
}

void stopMotors() {

```

```

    // Stop all motors
    digitalWrite(in1, LOW); digitalWrite(in2, LOW);
    digitalWrite(in3, LOW); digitalWrite(in4, LOW);
    digitalWrite(in5, LOW); digitalWrite(in6, LOW);
    digitalWrite(in7, LOW); digitalWrite(in8, LOW);
}

int measureDistance() {
    // Send a 10us pulse to trigger pin
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    // Measure echo time
    duration = pulseIn(echoPin, HIGH);

    // Convert to distance in cm
    return duration * 0.034 / 2;
}

```

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